Statistics & Probability

ANSWER KEY

Section 1: Summary Statistics & Outliers

1 The mean of five numbers is 24. When a sixth number is added, the mean becomes 26. What is the sixth number?

a. 32

b. 34

c. 36 ✓

d. 38

Sum of 5 numbers $= 5 \times 24 = 120$. Sum of 6 numbers $= 6 \times 26 = 156$. Sixth number $= 156 - 120 = 36$.

2 What is the median of the data set: $\{14, 8, 22, 19, 11, 25, 17, 30\}$?

a. 17

b. 18 ✓

c. 19

d. 20.5

Sort in ascending order: $8, 11, 14, 17, 19, 22, 25, 30$. The two middle numbers are 17 and 19. Median $= \frac{17+19}{2} = 18$.

3 An outlier is defined as any data point greater than $Q_3 + 1.5 \times \text{IQR}$. If $Q_1 = 20$ and $Q_3 = 35$, which value is considered an outlier?

a. 50

b. 55

c. 58

d. 60 ✓

$\text{IQR} = 35 - 20 = 15$. Upper threshold $= 35 + 1.5(15) = 35 + 22.5 = 57.5$. Since $60 > 57.5$, 60 is an outlier.

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Section 2: Normal Distributions & Standard Deviation

4 Exam scores are normally distributed with $\mu = 75$ and $\sigma = 6$. What percentage of scores fall between 69 and 81?

a. 50%

b. 68% ✓

c. 95%

d. 99.7%

The interval 69 to 81 represents $\mu \pm 1\sigma$ ($75 - 6$ to $75 + 6$). By the Empirical Rule, this contains approximately 68% of the distribution.

5 If a test score of 88 has a z-score of 2.5 with standard deviation $\sigma = 4$, what is the mean score μ?

a. 76

b. 78 ✓

c. 80

d. 82

$z = \frac{x - \mu}{\sigma} \Rightarrow 2.5 = \frac{88 - \mu}{4} \Rightarrow 10 = 88 - \mu \Rightarrow \mu = 78.$

6 Which change will increase the standard deviation of a dataset?

a. Adding 10 to all data points

b. **Multiplying all data points by** 2 ✓

c. Subtracting the mean from each data point

d. Dividing all data points by 3

Multiplying by 2 doubles the dispersion and doubles the standard deviation. Adding/subtracting a constant shifts the data without affecting standard deviation.

Section 3: Compound & Conditional Probability

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7 A fair six-sided die is rolled twice. What is the probability that both rolls result in a prime number?

a. $\frac{1}{4}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{1}{6}$

Prime numbers on a die are $\{2, 3, 5\}$ (3 out of 6, probability $\frac{1}{2}$). For two independent rolls: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

8 If $P(A) = 0.5$, $P(B) = 0.6$, and $P(A \cup B) = 0.8$, what is $P(A \cap B)$?

a. 0.1

b. 0.2

c. 0.3 ✓

d. 0.4

$P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.8 = 0.5 + 0.6 - P(A \cap B) \Rightarrow P(A \cap B) = 1.1 - 0.8 = 0.3$.

9 A box contains 5 red balls and 7 black balls. Two balls are drawn in succession without replacement. What is the probability that both balls are red?

a. $\frac{5}{33}$ ✓

b. $\frac{5}{36}$

c. $\frac{25}{144}$

d. $\frac{1}{6}$

Probability of first red $= \frac{5}{12}$. Probability of second red $= \frac{4}{11}$. Combined probability $= \frac{5}{12} \times \frac{4}{11} = \frac{5}{33}$.

Section 4: Permutations & Combinations

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10 How many distinct 3-person executive committees (President, Vice-President, Treasurer) can be elected from a club of 10 members?

a. 120

b. 240

c. 720 ✓

d. 1000

Since roles are distinct, order matters (permutation): $P(10, 3) = 10 \times 9 \times 8 = 720$.

11 How many subsets of 3 elements can be formed from a set of 8 elements?

a. 24

b. 56 ✓

c. 112

d. 336

Combinations: $\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$.

12 In how many ways can 4 math books and 3 science books be arranged on a shelf if all math books must be grouped together and all science books must be grouped together?

a. 144

b. 288 ✓

c. 576

d. 5040

There are $2!$ ways to arrange the two subject groups. Math books can be permuted in $4! = 24$ ways, and science books in $3! = 6$ ways. Total $= 2 \times 24 \times 6 = 288$.

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